

Customer Shopping Behavior Analysis

Data Analytics Project Report

1. Project Introduction

Overview

Customer behavior analysis helps businesses understand purchasing patterns, customer preferences, and factors influencing sales. This project focuses on analyzing customer shopping data to identify trends in purchasing habits, customer segments, product performance, and subscription behavior.

The analysis is performed using **Python, SQL, and Power BI** to transform raw customer transaction data into meaningful business insights.

Project Objective

The main objectives of this project are:

- Analyze customer purchasing patterns
 - Identify high-value customers
 - Understand product and category performance
 - Study the impact of discounts and subscriptions
 - Create a dashboard for business decision-making
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2. Project Workflow

The complete workflow followed in this project:

Data Collection



Data Cleaning & Preparation (Python)



Exploratory Data Analysis



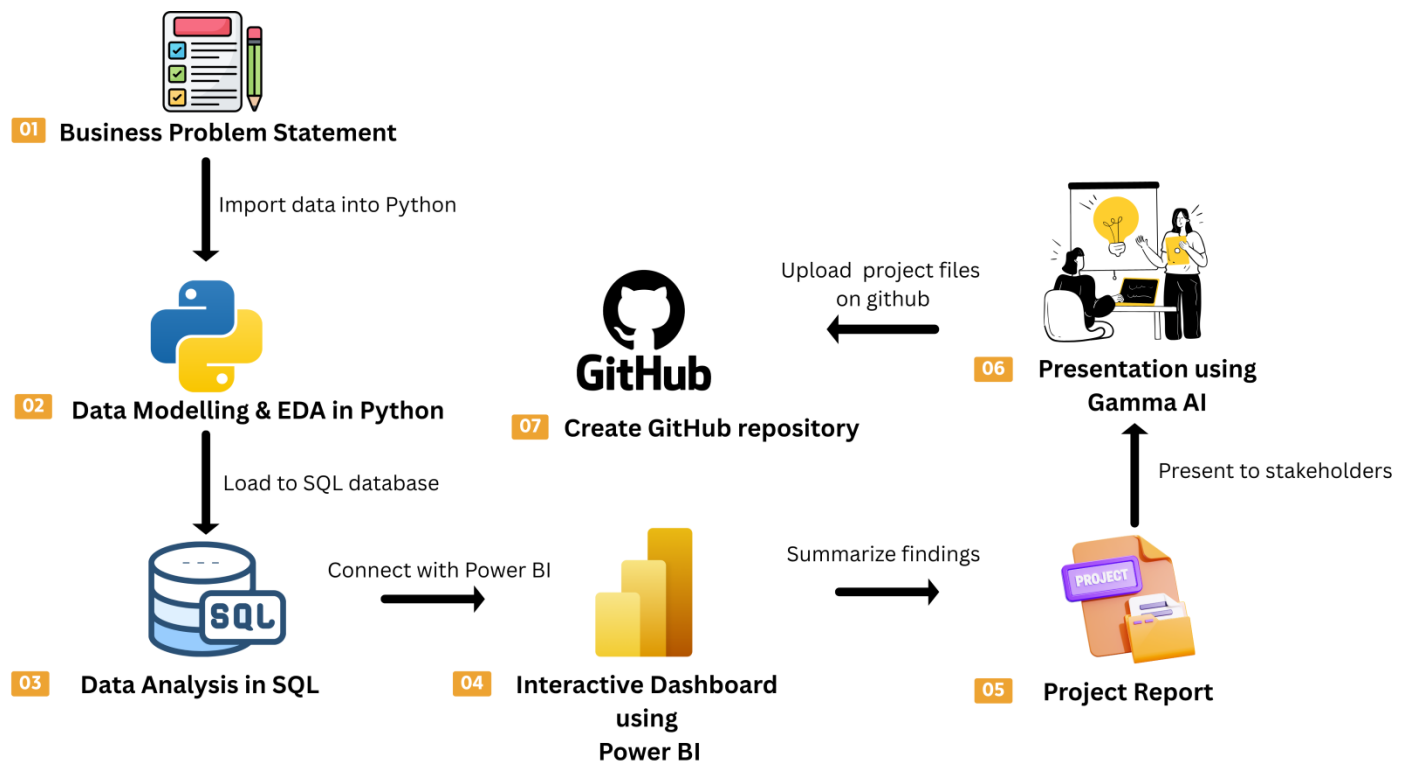
Business Analysis (SQL)



Dashboard Creation (Power BI)



Insights & Recommendations



3. Dataset Description

The dataset contains customer shopping information including customer details, purchase history, product information, and shopping preferences.

Dataset Overview

- Total Records: 3900
- Total Features: 18

Main Features:

Category	Features
Customer Details	Age, Gender, Location
Purchase Details	Item Purchased, Category, Purchase Amount
Shopping Behaviour	Discount, Previous Purchases, Frequency
Customer Feedback	Review Rating
Delivery Information	Shipping Type
Membership	Subscription Status

customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes
...	...	...	...	...	...	...	...	...	...	...	...	...
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No
3899	3900	52	Female	Handbag	Accessories	81	California	M	Beige	Spring	3.1	No
3900 rows × 19 columns												

4. Data Cleaning and Preparation

Before analysis, the dataset was cleaned and prepared to ensure accuracy.

Data Quality Checks

The following steps were performed:

- Checked dataset structure
- Reviewed missing values
- Identified incorrect data types
- Verified duplicate records

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Customer ID                          3900 non-null   int64
 1   Age                                  3900 non-null   int64
 2   Gender                              3900 non-null   object
 3   Item Purchased                       3900 non-null   object
 4   Category                            3900 non-null   object
 5   Purchase Amount (USD)                3900 non-null   int64
 6   Location                             3900 non-null   object
 7   Size                                 3900 non-null   object
 8   Color                                3900 non-null   object
 9   Season                               3900 non-null   object
10   Review Rating                        3863 non-null   float64
11   Subscription Status                  3900 non-null   object
12   Shipping Type                       3900 non-null   object
13   Discount Applied                    3900 non-null   object
14   Promo Code Used                     3900 non-null   object
15   Previous Purchases                   3900 non-null   int64
16   Payment Method                      3900 non-null   object
17   Frequency of Purchases               3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Missing Value Treatment

The dataset contained missing values in the review rating column.

These missing values were handled using category-based median values to maintain consistency.

```
df.isnull().sum()
```

```
Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64
```

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

Feature Engineering

New features were created to improve analysis:

Age Group

Customers were divided into different age groups to study buying behaviour across generations.

Example:

- Young Customers
- Middle Age Customers
- Senior Customers

Customer Segmentation

Customers were categorized based on purchasing history:

- New Customers
- Returning Customers
- Loyal Customers

```
frequency_mapping={
    'Fortnightly':14,
    'Weekly':7,
    'Monthly':30,
    'Quarterly':90,
    'Bi-Weekly':14,
    'Annually':365,
    'Every 3 Months':90
}
```

```
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)
```

```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

	purchase_frequency_days	frequency_of_purchases
0	14	Fortnightly
1	14	Fortnightly
2	7	Weekly
3	7	Weekly
4	365	Annually
5	7	Weekly
6	90	Quarterly
7	7	Weekly

5. Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand patterns and relationships in customer behaviour.

Customer Analysis

Questions explored:

- Which customer group spends the most?
- Which gender contributes more revenue?
- What is the distribution of customers?

```
(df['discount_applied'] == df['promo_code_used']).all()
```

```
True
```

```
df=df.drop('promo_code_used',axis=1)
```

```
df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'previous_purchases', 'payment_method',  
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
      dtype='object')
```

Product Analysis

Analysis performed:

- Top-selling products
- Best-rated products
- Category-wise performance

```
--Q1. What is the total revenue generated by male vs. female customers?
select gender, sum(purchase_amount) as revenue
from customer
group by gender

--Q2. Which customers used a discount but still spent more than the average purchase amount?
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes' and purchase_amount >= (select AVG(purchase_amount) from customer)

-- Q3. Which are the top 5 products with the highest average review rating?
select item_purchased, round(avg(review_rating::numeric),2) as "Average Product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5
```

Sales Behaviour Analysis

The impact of:

- Discounts
- Shipping type
- Purchase frequency

was studied to understand buying decisions.

```
--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.
select shipping_type,
ROUND(AVG(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;

--Q5. Do subscribed customers spend more? Compare average spend and total revenue
--between subscribers and non-subscribers.
SELECT subscription_status,
COUNT(customer_id) AS total_customers,
ROUND(AVG(purchase_amount),2) AS avg_spend,
ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

6. SQL Business Analysis

After cleaning the data, the dataset was loaded into a SQL database for business-level analysis.

The following questions were answered:

Revenue Analysis

- Revenue contribution by gender
- Revenue by age group

Customer Behaviour

- High-spending customers
- Repeat buyers
- Subscription behaviour

Product Insights

- Top products by rating
- Products frequently purchased with discounts

```
--Q6. Which 5 products have the highest percentage of purchases with discounts applied?
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

```
--Q7. Segment customers into New, Returning, and Loyal based on their total
-- number of previous purchases, and show the count of each segment.
with customer_type as (
SELECT customer_id, previous_purchases,
CASE
  WHEN previous_purchases = 1 THEN 'New'
  WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
  ELSE 'Loyal'
END AS customer_segment
FROM customer)
```

7. Power BI Dashboard

An interactive dashboard was created to present important findings visually.

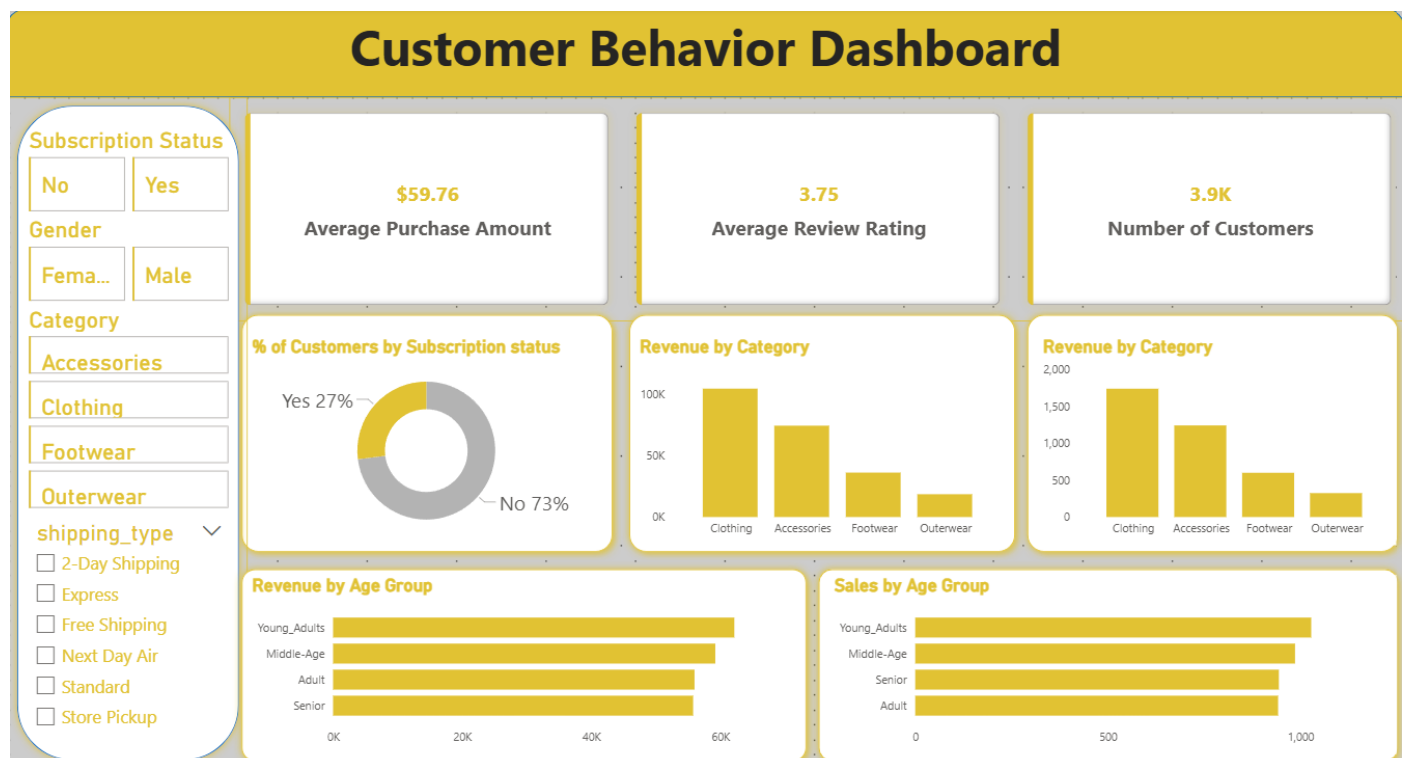
The dashboard contains:

Key Performance Indicators

- Total Revenue
- Average Purchase Amount
- Number of Customers
- Subscription Rate

Visualizations

- Sales by category
- Customer segmentation
- Product performance
- Purchase trends



8. Key Findings

Based on the analysis:

- Certain customer groups contribute significantly to revenue
- Repeat customers show stronger purchasing patterns
- Some products perform better in terms of ratings and sales
- Discounts influence customer purchase behaviour
- Subscription customers provide valuable business opportunities

9. Business Recommendations

Based on the insights:

Customer Loyalty Program

Businesses can introduce rewards for repeat customers to increase retention.

Personalized Marketing

Offers can be targeted based on customer age groups and purchase history.

Discount Optimization

Discount strategies should be balanced to increase sales while maintaining profitability.

Product Strategy

High-performing products can be promoted more aggressively.

10. Conclusion

This project demonstrates how data analytics can help businesses understand customer behaviour and make data-driven decisions.

By combining Python, SQL, and Power BI, raw customer data was transformed into actionable insights related to sales, customer loyalty, and product performance.